1.12 Classwork: Simpson's Paradox — Markscheme

Lesson problem set, not scored: model answers and teaching notes. Correlation descriptors follow the textbook scale: $|r| < 0.25$ very weak, 0.25–0.5 weak, 0.5–0.75 moderate, 0.75–0.95 strong. Data and figures: `claude-ib-project/output/correlation_claude/` *(data set D).*

1. **(a)** $100\,000 \times 1.0622 = 106\,220$ US dollars.

   (b) $\dfrac{106\,220}{1.5759} = 67\,402.75\ldots$, so EUR 67 400.

   $$\text{percentage change} = \frac{67\,402.75\ldots - 100\,000}{100\,000} \times 100 = -32.6\% \quad (-32.597\ldots)$$

   The dollar weakened by about a third against the euro. Common error: multiplying by 1.5759 *instead of dividing (USD → EUR divides by EUR/USD), which gives a "gain" to EUR* 167 000*.*

   (c) EUR/USD fell from 1.5759 to 1.2744, so the dollar became *stronger*.

   $\dfrac{106\,220}{1.2744} = 83\,349.02\ldots$, so EUR 83 300 (accept 83 349).

   Teaching note: in four months the same dollars gained about 24% *in euro terms (*83 349/67 403*). In the autumn 2008 crisis, investors worldwide bought dollars and US Treasuries as a safe haven.*

   (d) The yield fell from 4.01% to 2.42%, a fall of 1.59 percentage points. When yields fall, the prices of existing bonds rise, so this was good for the investor.

   Accept "fell steeply in December 2008". Late 2008 was good for the investor on both counts: bond prices up and the dollar stronger.

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2. **(a)** 2003–2007: $r = 0.39$, weak positive.

   2008–2011: $r = 0.46$, weak positive.

   2003–2011: $r = -0.30$, weak negative.

   Accept "moderate" for 0.46*. The point to draw out: both periods are positive, the whole is negative.*

   (b) From the first period to the second, the mean yield fell (4.40 → 3.23) and the mean EUR/USD rose (1.249 → 1.396). So the 2008–2011 cloud of points lies up and to the left of the 2003–2007 cloud. A line fitted to all the points mostly joins the two clouds, and it slopes down. This shift between the periods outweighs the upward slope inside each period.

   Teaching background (law of total covariance): the whole-period covariance is the weighted within-period covariance plus a between-period term $\sum_k w_k(\bar{x}_k - \bar{x})(\bar{y}_k - \bar{y})$*. Here the between-period term is negative and dominates.*

   (c) Within each period the relationship is the opposite: months with higher yields had a higher EUR/USD, a *weaker* dollar. The negative whole-period value comes from the difference between the two periods, not from how the series move together month by

month. Yields and the exchange rate both drifted over the years, so a correlation of *levels* mainly reflects those drifts and changes with the dates chosen.

Also accept: correlation does not show causation; a third factor (the 2008 crisis and the Federal Reserve's rate cuts) drove both.

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3. **(a)**

Month (2008)	Jul	Aug	Sep	Oct	Nov	Dec
change in yield (pp)		−0.12	−0.20	+0.12	−0.28	−1.11
change in EUR/USD (%)		−5.10	−4.10	−7.50	−3.93	+6.02

For example, September: $\dfrac{1.4342 - 1.4955}{1.4955} \times 100 = -4.0990\ldots$. *Common errors: percentage change of the yield* ($-0.20/3.89 = -5.14\%$) *instead of the change in percentage points; dividing by this month's value instead of last month's.*

(b) 2003–2007: $r = -0.25$; 2008–2011: $r = -0.15$; 2003–2011: $r = -0.18$. All three are weak or very weak and *negative*. Unlike the levels in 2(a), the changes give the same sign for each period and for the whole period: there is no Simpson's paradox.

(c) $r < 0$: in a month when the US yield rises, EUR/USD tends to fall, so the dollar tends to get *stronger*. A stronger dollar raises the euro value of the investor's dollars. This currency gain partly offsets the loss on the bonds, so the two risks partly cancel, and the investor's total risk is smaller than it would be if the two moved independently.

The offset is small: $r = -0.18$ *is very weak. Class examples from the table: October 2008 (yield* $+0.12$*, dollar* 7.5% *stronger) and December 2008 (yield* -1.11*, bond gain; dollar* 6% *weaker, currency loss).*

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4. **(a)** Period 1: $r = -1$. Period 2: $r = -1$. All six points: $r = 0.807$ $(0.80722\ldots)$

By hand: $\bar{x} = 4.5$, $\bar{y} = 6.5$, $S_{xy} = 33.5$, $S_{xx} = S_{yy} = 41.5$, *so* $r = 33.5/41.5$.

(b) Each period lies on a straight line with gradient -1, but period 2 lies up and to the right of period 1: the means are $(2, 4)$ and $(7, 9)$. Taken together, the points rise from the first group to the second, so r for all six points is positive.

(c) Simpson's paradox appears when the groups have different means, so the clouds sit in different places. Monthly changes average close to zero in every period, so the clouds of changes sit on top of each other. There is no shift between the periods to create a trend of its own, and the whole-period r reflects the month-to-month relationship found inside each period.

(d) Model answer: "Correlate the monthly changes (returns) of the two prices, not the prices themselves: prices trend, so the correlation of levels mostly measures shared trends and changes sign with the dates you choose. Report r for the returns over the whole period and in each half, to check that the relationship is stable."

Teaching note: for gold and oil, 2015–2025, the monthly returns give $r \approx -0.04$ *for the whole period and for each half.*